

RAUNAK YADAV

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🌐 linkedin.com/in/raunak7888 🏠 github.com/Raunak7888 🌐 Portfolio

SUMMARY

Full-stack developer with experience building and deploying web applications, backend systems, and cloud-native infrastructure. Worked across the stack using React, Next.js, TypeScript, Java, and Spring Boot, with hands-on experience in Docker, Kubernetes, and CI/CD automation. Built projects ranging from real-time communication platforms and AI-powered tools to distributed systems, taking them from design and development through deployment and operations.

TECHNICAL SKILLS

Languages: Java, TypeScript, Python, HCL, Shell/Bash
Backend: Spring Boot, JWT, OAuth2, Express.js, Flyway
Frontend: React, Next.js, Tailwind CSS, Zustand, GSAP, shadcn/ui
Databases: PostgreSQL, MySQL, Redis
DevOps & Cloud: Docker, Kubernetes, Terraform, Ansible, GitHub Actions, GCP, Prometheus, Grafana
AI & Tools: Google Gemini API, Claude, Maven, Gradle

PROJECTS

- Hermes** | *Spring Boot, React, Kubernetes, Terraform, Ansible, GCP* | 🏠 GitHub | 🌐 Live Jan 2025 – Present
- Provisioned a two-VM GCP Kubernetes cluster from scratch using Terraform for infrastructure and Ansible for node bootstrapping, then deployed a full-stack real-time chat platform across it with zero manual steps.
 - Implemented dual-path authentication supporting Google OAuth2 and email/OTP login, issuing JWT access and refresh tokens in HttpOnly cookies with per-user Redis sliding-window rate limiting and Redis image caching to reduce database load on hot paths.
 - Built a full CI/CD pipeline in GitHub Actions covering unit tests, Trivy security scans on Docker images and IaC, SHA-tagged multi-stage Docker builds, and automated kubectl rollout with smoke testing on every push.
 - Hardened the cluster with Kubernetes NetworkPolicies (default-deny), Traefik ingress with Let's Encrypt TLS, HPA for autoscaling, and Prometheus/Grafana dashboards across JVM, HTTP, WebSocket, and Redis metrics.
- AlgoForge** | *Express, Next.js, TypeScript, PostgreSQL, Redis, Gemini, pnpm Workspaces* | 🏠 GitHub May 2026 – Present
- Eliminated redundant AI provider calls for identical submissions by building a two-layer content-addressed cache (in-process LRU backed by Redis) keyed on SHA-256 of the code payload, serving repeat analyses without cost.
 - Abstracted AI providers behind a factory pattern supporting Google Gemini and OpenRouter interchangeably, enforcing strict JSON-contract prompting and Zod validation on every response before it reaches the application layer.
 - Built ForgeRecorder, a standalone TypeScript package covering 7 data-structure types and 18 highlight roles, recording deep-cloned step snapshots for frame-by-frame browser-side algorithm visualisation.
 - Enforced a single Zod schema across the API boundary in a pnpm monorepo, shared between Express validation and Next.js TypeScript types, and secured all mutation routes with CSRF double-submit cookies and custom SHA-256 session token hashing.
- Pokeverse** | *Spring Boot, Next.js, PostgreSQL, Redis* | 🏠 GitHub Aug 2025 – Dec 2025
- Designed a modular multiplayer game platform centralising authentication, room lifecycle, WebSocket event routing, and chat as reusable infrastructure, demonstrated with a real-time Pokemon Quiz supporting single-player and multiplayer modes.
 - Applied a dual-store architecture separating durable state (PostgreSQL with Flyway migrations) from transient live session data (Redis), reducing hot-path read latency during active gameplay.
- Calculate** | *Python, Algorithms, Data Structures* | 🏠 GitHub | 📝 Medium May 2025 – Jun 2025
- Implemented a compiler-inspired pipeline (tokenizer, implicit-multiplication injector, Shunting-Yard converter, stack evaluator) supporting 50+ functions and 70+ operators with zero external math dependencies, validated by 100+ test cases.
 - Published a technical article on Medium explaining the Shunting-Yard algorithm, operator precedence, and edge-case handling for a general engineering audience.

EDUCATION

DAV Institute of Management, Affiliated to MDU Rohtak Faridabad, Haryana
Bachelor of Computer Applications (BCA) Aug 2023 – Jun 2026

HIGHLIGHTS & CERTIFICATIONS

- Solved **300+** problems on LeetCode with focus on Trees, Graphs, and Dynamic Programming.
- Completed Java Spring Framework, Spring Boot and Spring AI Certification (Udemy).
- Earned Google Cloud Skill Badge: Build Real World AI Applications with Gemini.
- Earned Google Cloud Skill Badge: Prompt Design in Vertex AI.